

Jolly Roger

Trustless Cross-Chain RWA Graded Collectibles Powered by the Attestcoin Protocol (USC) on Creditcoin CC3

Hackathon: BUIDL CTC 2026 Fall	Track: Tokenized RWA & Cross-Chain Infra	Author: Kafir (@kafirdclxvi)
Live Demo: https://www.jollyroger.fun	Execution Chain: Creditcoin CC3 (102031)	Source Vault: Ethereum Sepolia (11155111)

1. Executive Summary & Core Thesis

High-value physical collectibles—primarily certified Gem Mint graded trading cards (PSA, BGS, and CGC slabs)—represent a rapidly expanding **\$20B+ real-world asset class**. However, bringing physical collectibles on-chain has historically suffered from fatal architectural vulnerabilities: reliance on centralized multisig bridge oracles, lack of cryptographic custody provenance, and cold, spreadsheet-like user experiences that alienate mainstream collectors.

Jolly Roger completely eliminates centralized bridge oracles by deploying the **Attestcoin Protocol (Universal Smart Contracts - USC)**. Physical cards are escrowed into custody on Ethereum Sepolia, and their custody state roots are directly proven on **Creditcoin CC3 Testnet** using the native **Block Prover Precompile (0x0FD2)**. Furthermore, Jolly Roger redefines Web3 onboarding through a tactile, gamified 2D collector sanctuary where 'The Scene IS the Product', enriched with live Fair Market Value (FMV) appraisal data from the **Renaiss Protocol API**.

Key Technological Innovation: Unlike conventional cross-chain bridges that depend on off-chain federated signers or relayer committees, Jolly Roger utilizes Creditcoin's native precompile (0x00000000000000000000000000000000FD2) to execute synchronous, mathematically verifiable Merkle Patricia inclusion proof validation directly inside the EVM.

2. The Problem: The Trilemma of Physical RWAs

Current physical asset tokenization systems suffer from three fundamental deficiencies:

- 1. The Oracle & Bridge Vulnerability:** Cross-chain protocols customarily rely on multisig bridge signers (e.g., 5-of-9 committees) or centralized oracle feeds to confirm physical vault deposits. Over \$3B has been stolen across DeFi due to compromised bridge keys and malicious oracles.
- 2. Disconnected Custody & Counterfeiting:** Physical custody receipts lack cryptographic binding to target execution layers. Slabs can be removed from physical vaults while their synthetic tokens continue circulating on-chain.
- 3. Broken User Experience & Misrouted Transactions:** Conventional Web3 platforms present cold, spreadsheet dashboards without network enforcement. Non-crypto-native collectors frequently send transactions on wrong networks (e.g., Arbitrum vs. Sepolia), resulting in lost assets.

3. Technical Architecture & Attestcoin (USC) Protocol

Jolly Roger operates across a trustless dual-chain infrastructure connecting Ethereum Sepolia and Creditcoin CC3 Testnet without third-party relay trust:

Step	Layer / Chain	Mechanism & Function Call	Cryptographic Security Guarantee
1. Physical Vault Escrow	Ethereum Sepolia (11155111)	vaultCard(certNumber, cardName, grade, appraisalUsd) in PhysicalVaultEscrow.sol	Locks card into custody; emits PhysicalCardVaulted with immutable parameters and unique keccak256 asset hash.
2. Proof Extraction	Attestcoin Proof Service	POST /proof { txHash, chainKey: 1 } via Attestcoin Proof Builder API	Generates Merkle Patricia Proof entries (siblings, roots) and block continuity lower endpoint digest.
3. Native Precompile Verification	Creditcoin CC3 (102031)	proveAndVaultCard(...) executes staticcall(0x0FD2, proofData)	Block Prover Precompile (0x0FD2) verifies Sepolia state root continuity directly at the protocol level.
4. Receipt Decoding & Validation	Creditcoin CC3 (102031)	EvmV1Decoder.decodeReceiptFields() at 0x731c...F9f	Confirms receiptStatus == 1, matches tx.to == sourceVaultAddress, and checks replay key.
5. Atomic RWA Minting	Creditcoin CC3 (102031)	CreditcoinRWAVaultASC.sol mints verified RWA token to collector	Token is cryptographically linked to physical slab with tamper-proof on-chain provenance.

4. Live Deployed Contracts & On-Chain Infrastructure

Both smart contracts and attestation components are deployed live on-chain and operational:

Network	Component / Contract	Verified Address / Endpoint	Explorer / Deployment Tx
Creditcoin CC3 Testnet (Chain ID: 102031)	CreditcoinRWAVaultASC (Attestcoin Smart Contract)	0x1a8757a621b0ac08aa91312e282307fb2e21b87f	Tx: 0x66a9d114...f7f0172b blockscout.com/address/0x1a87...
Ethereum Sepolia (Chain ID: 11155111)	PhysicalVaultEscrow (Source Physical Vault)	0x0068856c80535b518dbe2a10b56e3c25f9139bb4	Tx: 0xcd53d866...32c5956 sepolia.etherscan.io/address/0x0068...
Creditcoin CC3 Native	Block Prover Precompile	0x00FD2	Native protocol precompile (staticcall)
Creditcoin CC3 Verified	EvmV1Decoder	0x731c345d79Fb8BbDC541f9DF3b6317585F849F9f	Decodes transaction & receipt logs
Attestcoin Infrastructure	Proof Builder API	https://proof-builder.cc3-testnet.creditcoin.network	Live proof generation service

5. Interactive Features & 2D Collector Experience

Jolly Roger replaces cold, disconnected spreadsheets with a tactile 2D neon-lit gamer hideout. Every capability maps to an invariant furniture hotspot:

- **Smartphone Gate (phone):** Web3 wallet login modal featuring single-line Web3 pills (pulsing emerald network indicator, monospace address, avatar preview), session management, and on-chain Creditcoin Provenance ID (CTC-CC3- $\{address\}$) issuance.
- **Card Cabinet (cabinet):** Interactive physical collection shelf querying live Renaiss Protocol API. Features 3D perspective inspection, PSA/BGS grading certifications, and real-time Fair Market Valuations.
- **Cross-Chain Vault Escrow (escrow):** 3-step interactive custody vault: (1) Deposit slab into Sepolia escrow, (2) Generate Attestcoin Merkle proof, and (3) Execute CC3 precompile attestation with live progress trackers.
- **Chain-of-Custody Ledger (ledger):** Cryptographic ledger displaying attested slabs, block continuity hashes, and direct one-click links to Sepolia Etherscan and Creditcoin Blockscout.
- **Retro Arcade PC (computer):** Proof-of-play retro volleyball mini-game where collectors earn verified high scores and Creditcoin CC3 tCTC card reward drops.
- **Guestbook Note (note):** On-chain visitor guestbook supporting handwritten notes and direct tCTC tip gifting, guarded by NetworkGuard to prevent accidental cross-chain token loss.
- **Trophy Album (album):** Digital Soulbound Token (SBT) passport recording collector achievements and verified provenance stamps.
- **Time-of-Day Dynamic Theme:** Ambient lighting automatically transitions between bright daytime (06:00-17:59) and dark nighttime (18:00-05:59) based on the visitor's local system time.

6. Web3 Safety Guardrails & Cryptographic Security

Multi-chain applications are notoriously prone to user error. Jolly Roger implements defensive, multi-layered safeguards:

1. **Global NetworkGuard (components/ui/NetworkGuard.tsx):** Automatically detects when a connected wallet is on an unsupported chain (e.g., Arbitrum, Polygon, Ethereum Mainnet) and immediately prompts the wallet to switch to Creditcoin CC3 Testnet (102031).
2. **Transaction-Level Chain ID Pinning:** In both Guestbook tipping and Vault Escrow, transactions explicitly pass chainId: 102031 and chainId: 11155111 to Viem/Wagmi, halting execution immediately if the user rejects network switching.
3. **On-Chain Replay Attack Protection:** CreditcoinRWAVaultASC computes `queryKey = keccak256(abi.encodePacked(chainKey, blockHeight, transactionIndex))`. Once verified, `processedQueries[queryKey] = true` prevents duplicate minting or replayed proofs.
4. **Zero Mock Data Standard:** All slab grades, certifications, and high-resolution card artwork are pulled dynamically from live APIs (`api.renaiss.xyz/v0/`), ensuring real market valuation integrity.

7. Tech Stack & Future Roadmap

Technical Architecture: Built with Next.js 14 App Router, TypeScript, Tailwind CSS (Dark Base #0E0C17 + Neon Purple #B78CFF), Viem 2, Wagmi 2, RainbowKit, Solidity ^0.8.24, Supabase (PostgreSQL with RLS), Cloudflare R2, and Vercel.

Phase 1 (Hackathon Milestone - Completed): Dual-chain deployment on Ethereum Sepolia & Creditcoin CC3 Testnet, Attestcoin Block Prover Precompile (0x0FD2) integration, dynamic Renaiss Protocol API integration, and global NetworkGuard auto-switch protection.

Phase 2 (Mainnet Expansion): Production deployment on Creditcoin CC3 Mainnet in partnership with licensed physical vault custodians (PSA Vault, PWCC, Collectors Universe), enabling direct institutional custody intake.

Phase 3 (DeFi & Financialization): Collateralized lending against attested Gem Mint slabs, peer-to-peer decentralized slab swapping, and ERC-20 fractionalization pools for ultra-rare holy grail cards (e.g. Manga Luffy).

8. Solo Builder & Project Verification

Jolly Roger was architected and built from scratch by solo developer **Kafir (@kafirdclxvi / Kafiirr)**, encompassing dual-chain smart contract engineering, native Attestcoin precompile verification, Next.js frontend development, 2D scene orchestration, and custom Web3 UI/UX design.

By mathematically replacing centralized multisig bridge oracles with the **Attestcoin Protocol (USC)**, Jolly Roger sets a new benchmark for trustless physical Real-World Assets on Creditcoin CC3.

BUIDL CTC 2026 Fall Hackathon Verification Checklist	Status & Direct URLs
Track & Theme	Tokenized RWA & Cross-Chain Infrastructure — Attestcoin Protocol (USC)
Live Web Application	https://www.jollyroger.fun (Fully functional)
Open-Source GitHub Repository	https://github.com/Kafiirr/Jollyroger (Includes comprehensive README)
Creditcoin CC3 Smart Contract	0x1a8757a621b0ac08aa91312e282307fb2e21b87f (Precompile 0x0FD2 verified)
Ethereum Sepolia Escrow Contract	0x0068856c80535b518dbe2a10b56e3c25f9139bb4 (Live deployed vault)
Project Whitepaper (PDF URL)	https://www.jollyroger.fun/whitepaper.pdf